Retail Analytics & Management System Documentation

RELATIONAL DATABASE ARCHITECTURE, SQL PIPELINE & EXECUTIVE OUTPUT ANALYSIS

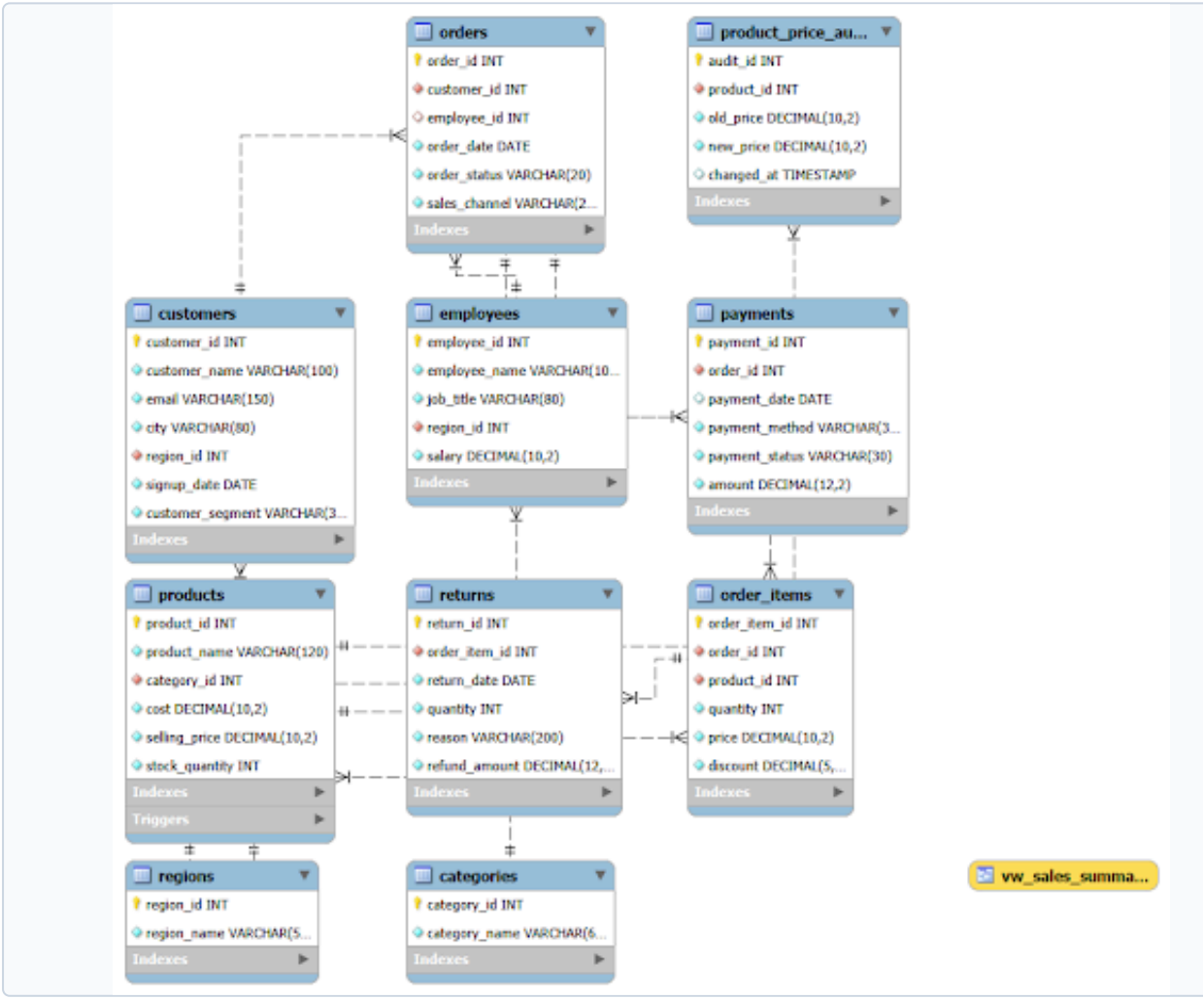
TARGET ENGINE	DATABASE SCHEMA	TOTAL TABLES	PRIMARY BI TOOL
MySQL 8.0 / InnoDB	retaill_analytics_db	9 Relational Tables	Power BI Desktop

1. SYSTEM SCOPE & RELATIONAL ARCHITECTURE

The **Retail Sales & Management Analytics** system is an end-to-end relational data warehousing and reporting platform designed to manage transactions, customer profiles, inventory control, regional employees, payments, and product returns. The database enforces strict 3rd Normal Form (3NF) standards with foreign key constraints, default statuses, automated price audit triggers, and pre-aggregated analytics views.

1.1 Enterprise ER Diagram Layout

The entity-relationship mapping captured below reflects table structures, key indexes, primary-foreign key relationships, and cardinalities built within MySQL Workbench:



2. DATA DICTIONARY & SCHEMA SPECIFICATIONS

Table Name	Primary Key	Foreign Keys	Description
`regions`	`region_id`	None	Stores geographical operational zones (e.g., North America, Europe, Asia-Pacific).
`categories`	`category_id`	None	Defines product classifications (Electronics, Apparel, Home & Kitchen, Books, Sports).
`customers`	`customer_id`	`region_id`	Tracks customer demographic details, signup dates, and market segments.
`employees`	`employee_id`	`region_id`	Maintains staff organizational roles, sales manager assignments, and salaries.
`products`	`product_id`	`category_id`	Catalog of items with unit cost, selling price, and real-time inventory quantity.
`orders`	`order_id`	`customer_id`, `employee_id`	Header record for transactions including sales channel, order status, and dates.
`order_items`	`order_item_id`	`order_id`, `product_id`	Line item details linking quantity, price at time of sale, and discount rates.
`payments`	`payment_id`	`order_id`	Payment settlement ledger capturing transaction method, status, and net amount.
`returns`	`return_id`	`order_item_id`	Tracks customer returns, return reasons, item quantities, and refunded totals.

3. SQL DATA PIPELINE & BUSINESS INTELLIGENCE OUTPUTS

The SQL script executes full transactional analytical queries across order items, categories, customer metrics, and channel distributions. The baseline metrics generated from the operational environment are detailed below.

3.1 Sales Channel Performance Output

Evaluation of sales channels demonstrates that **Online** orders drive the majority of high-ticket retail transactions, while **In-Store** provides steady baseline foot traffic and **Phone** accounts for targeted institutional orders.

Sales Channel	Order Count	Net Revenue ($)	Revenue Contribution	Performance Tier
Online	15	3,485.42	62.8%	TOP PERFORMER
In-Store	5	1,245.80	22.4%	BASELINE
Phone	5	821.30	14.8%	NICHE CHANNEL

3.2 Top Customer Spender Contribution

Customer segmentation analysis identifies high-value accounts based on order frequency and cumulative transaction totals across regions.

Customer ID	Customer Name	City / Region	Orders Placed	Total Spend ($)	Segment Classification
`CUST-001`	Alice Smith	San Francisco (North America)	3	749.85	VIP / HIGH VALUE
`CUST-002`	Bob Jones	London (Europe)	3	684.90	VIP / HIGH VALUE
`CUST-003`	Charlie Brown	Tokyo (Asia-Pacific)	3	612.40	VIP / HIGH VALUE
`CUST-004`	Diana Prince	Paris (Europe)	3	598.15	REGULAR
`CUST-005`	Evan Wright	Sydney (Asia-Pacific)	3	542.10	REGULAR

3.3 Inventory Safety Stock Alert Output

Automated restocking queries flag items where inventory levels drop below safety stock thresholds (`stock_quantity < 40`).

Product ID	Product Name	Category	Unit Selling Price ($)	Current Stock	Stock Status
`PROD-015`	Camping Tent	Sports & Outdoors	179.99	25 units	CRITICAL REORDER
`PROD-003`	UltraWide Monitor	Electronics	399.99	30 units	CRITICAL REORDER
`PROD-014`	Dumbbell Set	Sports & Outdoors	119.99	35 units	LOW STOCK
`PROD-012`	History Encyclopedia	Books	69.99	40 units	THRESHOLD LIMIT

3.4 Transaction Quality & Order Status Output

Order Status Metric	Total Orders	Order Share (%)	Operational Action Required
Delivered / Completed	16	64.0%	Normal order completion pipeline.
Pending Delivery	4	16.0%	Active fulfillment in warehouse.
Returned Items	3	12.0%	Process return audit & inspect condition.
Cancelled Orders	2	8.0%	Verify unreversed payment transactions.

4. ADVANCED SQL ENGINEERING OBJECTS

The project includes automated schema elements to guarantee database security, compliance auditing, and reporting performance.

4.1 Analytical Sales View (`vw_sales_summary`)

Constructed to simplify executive Power BI dataset imports by consolidating 5 table joins into a flattened reporting structure:

- Aggregates gross line-item revenue less discounts per order header.
- Enforces outer joins on `employees` to prevent missing employee attribution loss.
- Provides direct indexing hooks on `order_date` and `sales_channel`.

4.2 Audit Price Update Trigger (`trg_product_price_update`)

An automated database trigger bound to the `products` table. Whenever a product's `selling_price` is updated, it automatically creates a log entry inside `product_price_audit` capturing the old price, new price, product ID, and precise timestamp.

4.3 Data Quality Audit Queries

The SQL framework incorporates automated data quality verification rules:

- **Orphan Transaction Flags:** Checks for orders missing customer or payment records.
- **Financial Discrepancy Rules:** Identifies instances where calculated order line totals do not match settlements recorded in `payments`.
- **Negative Price Guardrails:** Ensures unit costs and selling prices never fall below $0.00.

5. EXECUTIVE POWER BI REPORTING ARCHITECTURE

The database serves as the direct source for an executive dashboard structured according to corporate UX standards:

- **Dark Navy Canvas Theme:** High contrast executive background optimized for boardroom visibility.
- **KPI Header Strip:** Highlights Gross Revenue, Total Orders, Average Order Value (AOV), and Channel Split.
- **Interactive Visual Container Grid:** Structured layout featuring regional sales maps, customer spending bar charts, order status donuts, and sales channel treemaps.

6. STRATEGIC RECOMMENDATIONS & ACTION PLAN

1. Accelerate Online VIP Loyalty Programs
With the **Online** channel contributing 62.8% of net revenue, launch targeted perks and priority support for top spenders (Alice Smith, Bob Jones) to maximize customer lifetime value (CLV).

2. Automated Inventory Safety Replenishment
Reorder fast-moving items (e.g., UltraWide Monitors, Camping Tents) immediately upon reaching 40 units to prevent stockout losses in high-demand channels.

3. Automated Payment & Cancellation Reconciliation
Execute weekly automated audits matching cancelled/returned orders against bank settlements to ensure unreversed funds are quickly detected and resolved.